NETWORK SECURITY LABORATORY

Experiment 1 – Introduction to Network Security Lab Tools and Simulators

Department: Cybersecurity Engineering Technologies | Fourth Year

Item	Description
Duration	2 Hours
Main Software	Cisco Packet Tracer
Additional Tools	Windows Command Prompt and Wireshark
Difficulty	Beginner

1. Main Objectives

- Recognize the basic tools used in a network-security laboratory.
- Create a simple Local Area Network (LAN) using Cisco Packet Tracer.
- Configure Internet Protocol Version 4 (IPv4)addresses and test connectivity using ping.
- Observe Address Resolution Protocol (ARP) and Internet Control Message Protocol (ICMP) packets in Simulation Mode.
- Use basic Windows network commands and become familiar with Wireshark.

2. Main Tools

Tool	Main Purpose
Cisco Packet Tracer	Build, configure and simulate networks.
Command Prompt	Run basic network commands such as ipconfig and ping.
Wireshark	Capture and inspect network packets.

3. Laboratory Safety

- Use tools only on laboratory systems and networks.
- Do not scan, attack or access another device without permission.
- Use cybersecurity tools only for educational and authorized work.

Key idea: Authorization is required before any network-security test.

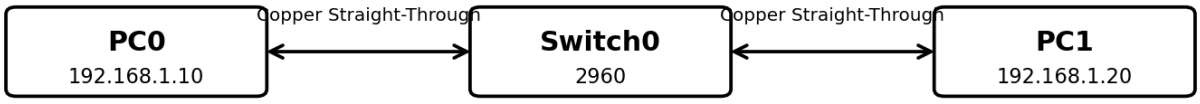
4. Basic Windows Network Commands

Command	Purpose
hostname	Displays the computer name.
ipconfig /all	Displays IP address, subnet mask, gateway and adapter details.
ping 127.0.0.1	Tests the local TCP/IP stack (loopback).
arp -a	Displays known IP-to-MAC mappings.
netstat -ano	Displays active network connections and related process IDs.

5. Build the Packet Tracer Network

1. Add two PCs and one Cisco 2960 switch.
2. Connect both PCs to the switch using Copper Straight-Through cables.
3. Wait until the link indicators become green.

Simple LAN Topology



6. Configure IP Addresses

Device	IP Address	Subnet Mask	Default Gateway
PC0	192.168.1.10	255.255.255.0	Leave empty
PC1	192.168.1.20	255.255.255.0	Leave empty

Why no router? Both PCs are in the same IP network (192.168.1.0/24), so they communicate through the switch directly.

7. Test Connectivity

From PC0:

ping 192.168.1.20

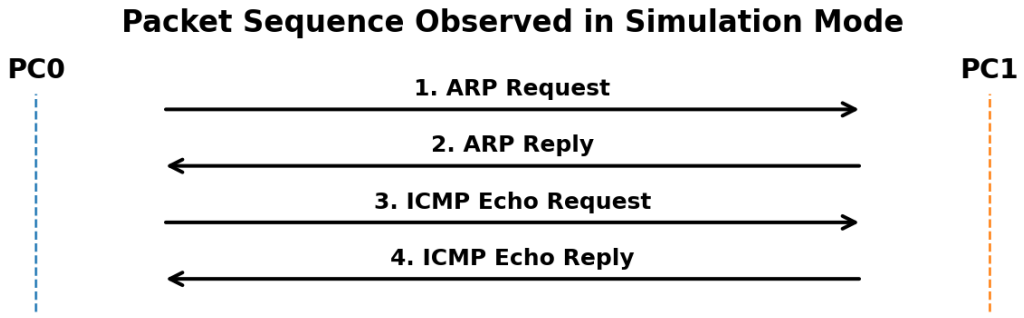
From PC1:

ping 192.168.1.10

Expected result: Successful replies. The first ping may fail while ARP learns the destination MAC address.

8. Observe Packets in Simulation Mode

1. Select Simulation Mode.
2. Open Edit Filters and keep only ARP and ICMP.
3. Run ping from PC0 to PC1.
4. Use Capture/Forward to watch each packet step by step.



Protocol	Purpose
ARP	Finds the MAC address associated with an IPv4 address.
ICMP	Used by ping to send Echo Request and Echo Reply messages.

9. Quick Troubleshooting Test

Change PC1 temporarily to 192.168.2.20/24. Ping from PC0 should fail because the two PCs are now in different IP networks and no router exists. Restore PC1 to 192.168.1.20 and test again.

10. Short Wireshark Introduction

1. Open Wireshark and select the active Ethernet or Wi-Fi interface.
2. Capture traffic for a short time, then press Stop.
3. Use simple display filters to focus on specific packets.

Filter	Shows
icmp	ICMP / ping packets
arp	ARP packets
dns	DNS packets
ip.addr == 192.168.1.10	Traffic related to PC0

11. Student Practical Task

1. Create a network with two PCs and one switch.
2. Configure PC0 = 192.168.10.10 and PC1 = 192.168.10.20 with mask 255.255.255.0.
3. Verify connectivity using ping.
4. Use Simulation Mode to observe ARP and ICMP.
5. Save the Packet Tracer project as Week1_StudentName.pkt.
6. Submit screenshots of the topology, IP configuration and successful ping.

12. Review Questions

Q: What is Cisco Packet Tracer used for? A: To create and simulate computer networks without physical equipment.

Q: What is the purpose of ping? A: To test whether a destination device is reachable.

Q: Which protocol is used by ping? A: ICMP.

Q: What is the purpose of ARP? A: To find the MAC address associated with an IPv4 address.

Q: Why is a switch used? A: To connect devices inside the same local area network.

Q: Why is a router not required here? A: Because both PCs are in the same IP network.

Expected Result

- Correct two-PC/switch topology
- Correct IPv4 settings
- Successful ping
- Visible ARP and ICMP packets
- Basic familiarity with Packet Tracer, Wireshark and Windows commands